Status: Not ready still under review. **INTRODUCTION TO QUANTUM COMPUTING**
IIT MADRAS & IBM RESEARCH INDIA

Week 3 **Assignment 3 Soln** **20 Marks** **Feb 2026**

- **This assignment has 10 questions of the multiple choice type.**
- **Each question can have multiple correct options. If you indicate *all* the correct answers, you get 2 marks.**
- **No marks will be awarded if you indicate an incorrect answer.**
- **If you indicate some, but not all, correct answers, you get 1 mark.**

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1. Consider the N-qubit computational basis state

$$|X\rangle = |X_{N-1}X_{N-2}\dots X_1X_0\rangle.$$

We are given

$$|\psi\rangle = \frac{1}{\sqrt{2^N}} \sum_{Y\in\{0,1\}^N} (-1)^{X\cdot Y}|Y\rangle,$$

where the mod-2 inner product is

$$X\cdot Y = (X_{N-1}Y_{N-1}) \oplus (X_{N-2}Y_{N-2}) \oplus \cdots \oplus (X_0Y_0).$$

Step 1: Action of the Hadamard Transform

Recall the single-qubit Hadamard action:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

For N qubits, the tensor product gives the identity:

$$H^{\otimes N}|X\rangle = \frac{1}{\sqrt{2^N}} \sum_{Y\in\{0,1\}^N} (-1)^{X\cdot Y}|Y\rangle.$$

Comparing with the given expression, we see immediately that

$$|\psi\rangle = H^{\otimes N}|X\rangle.$$

Hence the required unitary is

$$\boxed{\hat{U} = H^{\otimes N}.}$$

Step 2: Recovering X

Since the Hadamard transform is its own inverse,

$$H^{\otimes N}H^{\otimes N} = I,$$

applying $H^{\otimes N}$ again gives

$$H^{\otimes N}|\psi\rangle = H^{\otimes N} H^{\otimes N}|X\rangle = |X\rangle.$$

Measuring in the computational basis then yields X with probability 1.

Final Answer:

$$\boxed{\text{(c) } \hat{U} = H^{\otimes N}, \text{ and } X \text{ can be perfectly recovered.}}$$

2. In the Deutsch–Jozsa problem, a function $f : \{0,1\}^n \to \{0,1\}$ is promised to be either constant or balanced.

   To determine with 100% certainty whether the function is constant or balanced using a classical deterministic algorithm:

   Worst-case analysis:

   A balanced function outputs 0 for exactly 2^{n-1} inputs and 1 for the other 2^{n-1} inputs.

   In the worst case, the algorithm may query 2^{n-1} inputs and obtain the same output each time. At this stage, the function could still be constant, or it could be balanced (with the opposite value appearing in the remaining inputs).

   Therefore, one additional query is required to break the ambiguity.

   Hence, the minimum number of queries required is

$$2^{n-1} + 1.$$

   Final Answer:

$$\boxed{(c)\ 2^{n-1} + 1.}$$

3. Refer to the circuit shown in Figure 1

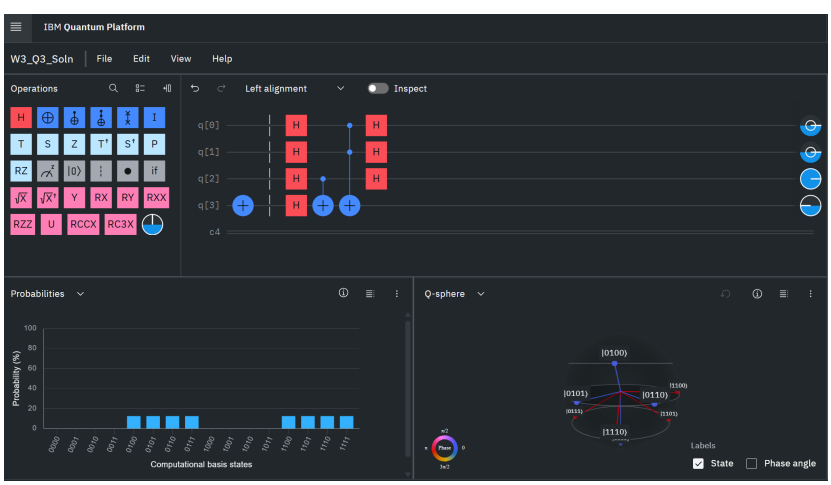


Figure 1: Circuit for W3_Q3

Final Answer:

$$\boxed{(c)\ |010\rangle \quad \text{and} \quad (d)\ |110\rangle.}$$

4. Both the Deutsch–Jozsa and Bernstein–Vazirani algorithms encode the function output $f(x)$ into the *phase* of the quantum state.

   Key properties exploited:

   - **Superposition:** The input register is prepared in an equal superposition of all possible x, allowing the function to be evaluated on all inputs simultaneously.
   - **Phase Kickback:** Using a suitably prepared ancilla qubit, the function value $f(x)$ is transferred into the phase factor $(-1)^{f(x)}$ of the computational basis states.

   The Uncertainty Principle is not directly used in this encoding process.

   Final Answer:

$$\boxed{(b)\ \text{Phase Kickback} \quad \text{and} \quad (c)\ \text{Superposition.}}$$

5. Refer to the circuit shown in Figure 2

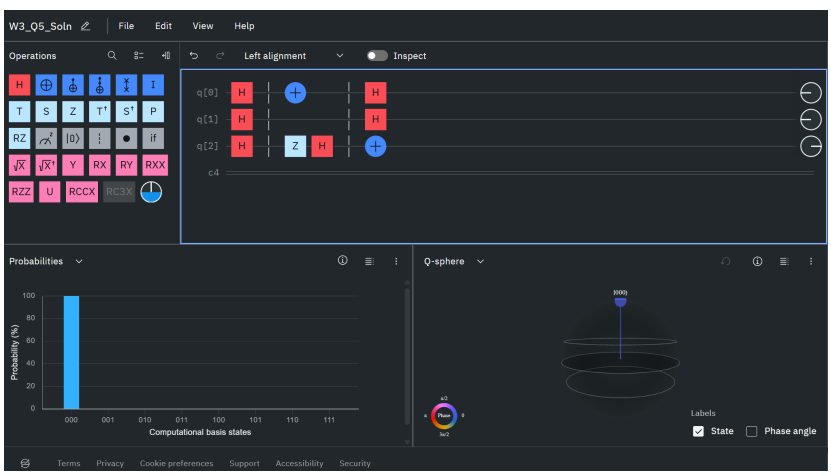
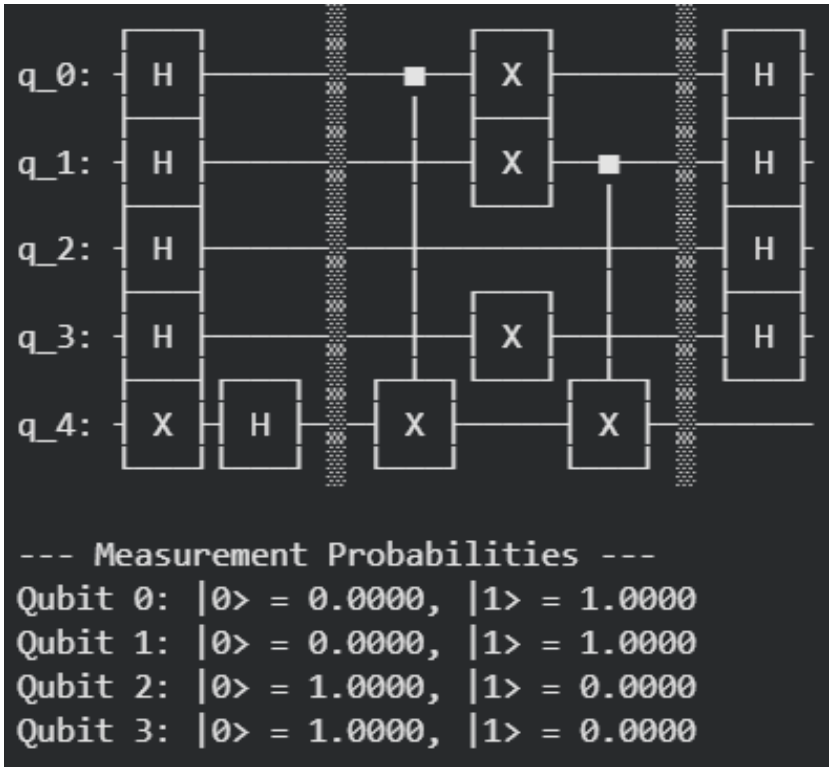


Figure 2: Circuit for W3_Q5

Final Answer:

$$\boxed{(d)\ |000\rangle.}$$

6. Refer to the circuit shown below (see `SolutionData2026/W3_Q6.png`).



From the circuit, we observe that not all four wires collapse to 0 at measurement.

Final Answer:

$$\boxed{(c)\ 0.}$$

7. The Bernstein–Vazirani algorithm uses the same overall circuit structure as the Deutsch–Jozsa algorithm.

   Both circuits:

   - Prepare the input register in superposition using Hadamard gates,
   - Apply an oracle that encodes the function into the phase,
   - Apply Hadamard gates again before measurement.

   The only difference lies in the specific function implemented by the oracle.

   Final Answer:

$$\boxed{(d)\ \text{It is identical in structure, differing only in the function implemented by the oracle.}}$$

8. The reflection about the uniform superposition state $|u\rangle$ (Grover diffuser) is constructed using Hadamard transforms.

   Construction:

   - Apply $H^{\otimes n}$ to transform $|u\rangle$ into $|0\ldots0\rangle$,
   - Perform a phase flip (reflection) about $|0\ldots0\rangle$,
   - Apply $H^{\otimes n}$ again to transform back.

   This effectively implements the reflection about $|u\rangle$.

   Final Answer:

   $\boxed{(a) \text{ By transforming } |u\rangle \text{ to } |0\ldots0\rangle \text{ with Hadamard gates, reflecting about } |0\ldots0\rangle, \text{ and transforming bac}}$

9. Refer to the circuit shown in Figure 3

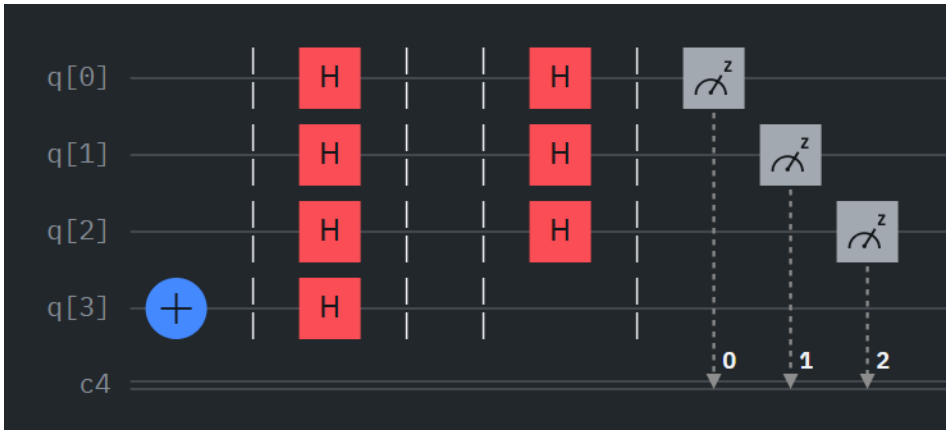


Figure 3: Circuit for W3_Q9

Step 1: Structure of the circuit

From Figure 3, we observe:

- The first three qubits have Hadamard gates applied at the beginning.
- The ancilla qubit is prepared in the $|-\rangle$ state.
- The oracle region (between dashed lines) does not apply any controlled operations depending on the input.
- A second layer of Hadamard gates is applied to the first three qubits.
- The first three qubits are then measured.

Step 2: Interpretation

Since the oracle does not introduce any input-dependent phase change, the function implemented is constant. In the Deutsch–Jozsa algorithm, this results in the final measurement being $|000\rangle$.

Final Answer:

$$\boxed{(a) \text{ Deutsch–Jozsa with Constant Function Oracle}}$$

10. In Grover's algorithm, the state evolution can be understood as a rotation in a two-dimensional subspace of the full Hilbert space.

    Geometric Interpretation:

    The relevant 2D plane is spanned by:

    - The target state $|a\rangle$,
    - The state $|e\rangle$, which is the uniform superposition of all non-target states.

    These two states are orthogonal and form a basis for the subspace in which Grover's iterations perform successive rotations toward $|a\rangle$.

    Final Answer:

    $\boxed{(b) \text{ The target state } |a\rangle \text{ and a uniform superposition of all non-target states, } |e\rangle.}$